

# Lanhao Niu

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## EDUCATION

- **University of Edinburgh** 09/2025 – 08/2026  
*MSc Cognitive Science* Edinburgh, UK
  - **Relevant Courses:** Accelerated Natural Language Processing; Text Technologies for Data Science; Machine Learning Systems; Seminar in Cognitive Modelling; Research Methods in Developmental Linguistics
- **Nanjing Agricultural University** 09/2021 – 06/2025  
*B.Eng. in Data Science and Big Data Technology* Nanjing, China
  - **Honours & Awards:** First Prize, Chinese Collegiate Computing Competition (2024); Grand Prize, Jiangsu Province Collegiate Computing Competition (2024)

## PUBLICATIONS

C=CONFERENCE, S=IN SUBMISSION

- [S.1] Lanhao Niu, Siddharth N. (2026). **LLMCMC: Language Models as Metropolis–Hastings Kernels for Probabilistic Program Inference**.  
Under review at *NeurIPS 2026 Workshop on Principles of Generative Modeling*.
- [C.1] Zhimin Hu, Lanhao Niu, Sashank Varma (2026). **The Representational Geometry of Number**.  
Accepted by *CogSci 2026*. arXiv:2602.06843 [cs.CL, cs.AI].
- [C.2] Zhimin Hu, Lanhao Niu, Sashank Varma (2026). **Language Models Represent and Transform Concepts with Shared Geometry**. Accepted by *ICML 2026 Mech Interp Workshop* (Spotlight 23/800).

## RESEARCH PROJECTS

- **LLMCMC: Language Models as MCMC Proposal Kernels** 01/2026 – 08/2026  
*Advisor: Prof. Siddharth Narayanaswamy, University of Edinburgh*
  - Proposed *LLMCMC*, an amortized probabilistic inference framework that uses autoregressive LLMs as state-conditioned Metropolis–Hastings proposal generators over probabilistic program text, observed data, and the current chain state, paired with a learned acceptance model.
  - Fine-tuned Qwen3.5-9B with LoRA on accepted NUTS transitions and evaluated zero-shot, in-context, and supervised fine-tuning regimes across 30 PosteriorDB programs, with component-level ablations and variable-name permutation tests to distinguish geometric learning from surface memorization.
  - Achieved near-NUTS in-distribution mixing (median ESS 94.4; normalized W1 0.195) and found that out-of-distribution chains can remain highly efficient while becoming distributionally misaligned, with failures concentrated on unseen combinations of program structure.
- **Representational Geometry of Concepts in LLMs** 08/2025 – 05/2026  
*Advisor: Prof. Sashank Varma, Georgia Institute of Technology*
  - Studied how concepts, numbers, and context-dependent meanings are geometrically organized in LLM hidden representations, and whether relational structure is preserved across different model architectures.
  - Developed cross-model analyses based on representational similarity, alignment, and transformation geometry to quantify shared structure in concept spaces and context-induced changes in representation.
  - Found stable relational geometry across architectures, including shared structure in contextual transformations and task-dependent representations; contributed to two accepted papers at CogSci 2026 and the ICML 2026 Mechanistic Interpretability Workshop (Spotlight).

## WORK EXPERIENCE

- **Z.AI (Zhipu AI)**  05/2024 – 12/2024  
*Research Intern* Beijing, China
  - Developed an end-to-end knowledge-graph Q&A system combining NER, Cypher query generation, and LLM-based reasoning for natural-language querying over structured enterprise knowledge bases.
  - Fine-tuned LLMs with LoRA for entity–relation extraction and natural-language-to-SQL generation, achieving over 90% accuracy; also contributed to RAG-based enterprise search and document-understanding pipelines.
- **DeepLang AI**  04/2024 – 05/2024  
*Data Engineering Intern* Beijing, China
  - Built data-processing pipelines for LLM pre-training and SFT dataset construction, including multi-format normalization, key-value extraction, NER-based annotation, and token-level preprocessing.

## SKILLS

- **LLM:** Supervised Fine-Tuning (LoRA), Representation Analysis, Prompting, RAG
- **ML:** PyTorch, HuggingFace Transformers, Probabilistic Programming, MCMC, Representation Learning
- **Engineering:** Python, C++ , Git, Docker, Google Cloud, Linux/Bash